When we talk about individual miners and the entire network size, we use different SI-Prefixes which can be confusing.

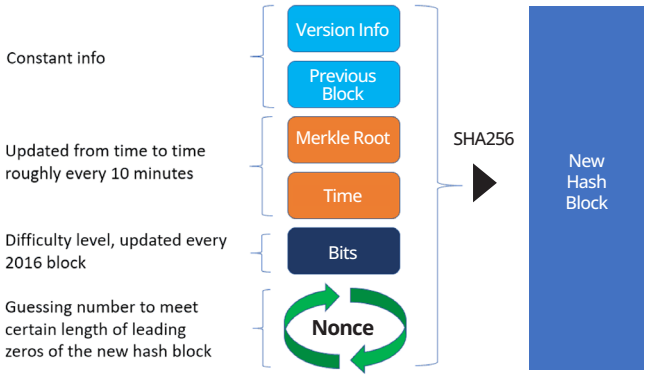
The main hash denotations are as follows:

- ***Bitcoin*** **mining machines** hash in Terahash per second (**TH/s**).
- The network's **total hash rate** is described in Exahash per second (**EH/s**)

Bitcoin Hash Rate:
1.1 Exahash / second

One hash / second
One **Kilo**hash = 1,000 hashes
One **Mega**hash = 1,000,000 hashes
One **Giga**hash = 1,000,000,000 hashes
One **Tera**hash = 1,000,000,000,000 hashes
One **Peta**hash = 1,000,000,000,000,000 hashes
One **Exa**hash = 1,000,000,000,000,000,000 hashes

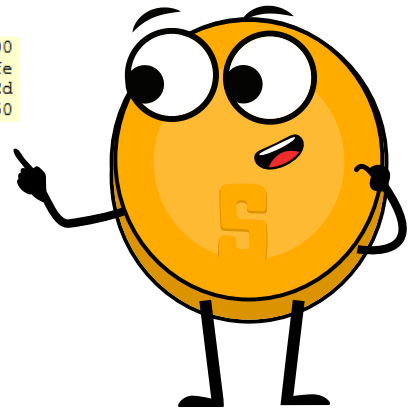
Bitcoin Block Hashing



The Block Hash can be referred to as the Proof of Work

In summary, every ten minutes or so, miners enter into a race to find a valid block hash. They begin taking all the data from their candidate blocks (which is summarized neatly in the block headers), double hashing it into a single hash, and comparing the output to a target hash value set by the network. If the block hash produced is too high, the miner adjusts the nonce and tries again, repeating this process trillions of times per second until one lucky miner finally finds a hash that meets the network's target.

version	02000000
previous block hash (reversed)	17975b97c18ed1f7e255adf297599b55 330edab87803c8170100000000000000
Merkle root (reversed)	8a97295a2747b4f1a0b3948df3990344 c0e19fa6b2b92b3a19c8e6badc141787
timestamp	358b0553
bits	535f0119
nonce	48750833
transaction count	63
coinbase transaction	
transaction	
...	

Block Hash

0000000000000000
e067a478024addfe
cdc93628978aa52d
91fabd4292982a50